

Andrew Ng Digital Twin

System Architecture & Design Decisions

Technical Report · June 2026

EXECUTIVE SUMMARY

This report describes the architecture, design decisions, and implementation strategy for a **Digital Twin of Andrew Ng**, an AI system engineered to emulate his knowledge, teaching methodology, reasoning process, communication style, and conversational behavior. The twin combines Retrieval-Augmented Generation (RAG), short and long term memory, persona modelling, and timeline-aware reasoning to deliver a conversational experience grounded in reliable source material.

CORE PRINCIPLES

The system is built on four core principles:

Accuracy ensures that every response is grounded in Andrew Ng's verified public work and source material.

Persona Consistency ensures that communication reflects Andrew Ng's structured, encouraging, and intuition-first teaching style.

Memory enables the system to retain both short-term conversational context and long-term user preferences, creating more personalized interactions.

Transparency ensures that all responses can be traced back to specific retrieved sources, providing clear auditability and trustworthiness.

SYSTEM ARCHITECTURE

The system operates through a **nine-stage pipeline** that transforms a raw user query into a retrieval-grounded, persona-consistent response while continuously maintaining and updating its memory store.

The pipeline consists of:

- User Query
- Memory Retrieval
- Knowledge Retrieval (RAG)
- Re-ranking
- Context Builder
- Persona Engine

- LLM (Gemini)
- Response Generation
- Memory Update.

Each stage is independently optimisable, allowing targeted improvements without requiring changes to the entire architecture. This modular design ensures that enhancements in retrieval quality, re-ranking precision, contextual understanding, or persona fidelity compound together, resulting in measurably better end-to-end response quality, consistency, and user experience.

KNOWLEDGE BASE CONSTRUCTION

The knowledge base is constructed from a diverse corpus of Andrew Ng's public work, ensuring broad coverage of his intellectual output across teaching, research, and industry commentary. Source categories include:

Stanford CS229 Lectures	Core machine learning theory, derivations, and pedagogical style.
DeepLearning.AI Content	Applied deep learning curriculum and practitioner guidance.
Coursera Course Materials	Structured educational content with exercises and explanations.
Interviews & Podcasts	Conversational tone, opinions, and forward-looking perspectives.
Conference Talks	Keynotes and panel discussions covering AI trends and strategy.
Blog Posts & Essays	Short-form analysis, industry commentary, and opinion pieces.
Research Papers	Technical depth, methodology, and academic rigor.

All documents are cleaned, normalized, deduplicated, and enriched with structured metadata (source type, publication date, topic tags) to support retrieval precision and timeline-aware reasoning.

CHUNKING STRATEGY

Parameter	Value	Purpose
Chunk Size	~900 tokens	Balances context richness against retrieval granularity.
Overlap	200 tokens	Preserves continuity across chunk boundaries.
Metadata	Full	Source, date, topic tags retained per chunk.

RETRIEVAL-AUGMENTED GENERATION (RAG)

The retrieval pipeline employs a hybrid approach combining the complementary strengths of dense vector search and sparse BM25 keyword search. This dual-mode strategy ensures that semantically related content and exact-match technical terminology are both surfaced effectively.

Dense Vector Search: Embeds queries and document chunks into a shared semantic space. Excels at capturing conceptual similarity and paraphrase matching, even when surface-level wording differs.

BM25 Keyword Search: Sparse lexical retrieval based on term frequency and inverse document frequency. Reliably surfaces exact-match results for technical terms, acronyms, and proper nouns.

Cross-Encoder Reranking: A neural cross-encoder re-scores the top candidates from the hybrid pool by jointly encoding the query and each document. This second-stage ranking substantially improves precision by leveraging full query–document interaction rather than independent embeddings.

PERSONA MODELLING

Persona modelling is the layer that transforms factually retrieved content into responses that genuinely reflect Andrew Ng's voice. The persona is derived from systematic analysis of his lectures, interviews, and public appearances, capturing the following characteristics:

Structured Exposition	Ideas are introduced with clear motivation, developed step-by-step, and concluded with a practical takeaway — mirroring his lecture structure.
Encouraging Tone	Responses are optimistic and supportive, particularly when addressing learners' doubts or misconceptions about difficulty.
Intuition-First Teaching	Complex mathematical or algorithmic concepts are preceded by an intuitive explanation before formal notation is introduced.
Engineering Pragmatism	Theory is consistently anchored to real-world applications, implementation considerations, and practical constraints.
Measured Confidence	Opinions are stated with appropriate epistemic humility; speculation is labelled as such.

MEMORY SYSTEM

The memory system operates across two temporal scopes to maintain both immediate conversational coherence and durable user-specific context:

Memory Type	Scope	Function	Persistence
Short-Term Memory	Current Session	Maintains turn-by-turn coherence; tracks topic threads and unresolved questions within a conversation.	Session duration only
Long-Term Memory	Cross-Session	Stores meaningful user facts, stated preferences, learning goals, and prior interactions for continuity.	Persistent across sessions

TIMELINE-AWARE REASONING

The system tracks temporal information embedded in source documents, including publication dates, historical viewpoints, and references to technological milestones, and uses this metadata to calibrate

responses to the appropriate point in time. This prevents unusual responses such as attributing current deep learning capabilities to early-period lectures, or treating superseded viewpoints as current positions.

Timeline-aware reasoning is particularly important when the user's query spans multiple periods of Andrew Ng's career, requiring the system to synthesise a temporally coherent narrative rather than conflating distinct positions held at different times.

PROMPT CONSTRUCTION

Final prompts are assembled from six ordered components, each contributing a distinct type of context required for a complete, accurate, persona-consistent response:

1. System Instructions	Establishes behavioural guidelines, output format, and safety constraints.
2. Persona Information	Injects the Andrew Ng persona profile, communication norms, and style rules.
3. Retrieved Knowledge	Top-ranked RAG results relevant to the user's current query.
4. Retrieved Memories	Applicable long-term and short-term memories for the current user and session.
5. Conversation History	Recent turn history to maintain thread coherence.
6. Current User Query	The user's verbatim input, positioned last to maximise attention weighting.

HALLUCINATION MITIGATION

Factual accuracy is maintained through a layered mitigation strategy rather than relying solely on the language model's parametric knowledge:

- **Retrieval Grounding:** All substantive claims are anchored to retrieved source documents, reducing reliance on model memory.
- **Source Prioritization:** Sources are ranked by authority and recency; higher-quality sources receive greater prompt weight.
- **Cross-Encoder Reranking:** Ensures the most relevant and accurate documents reach the context window.
- **Confidence-Aware Responses:** When retrieved evidence is insufficient, the system explicitly communicates uncertainty rather than confabulating.

DESIGN TRADE-OFFS

Each architectural choice reflects a deliberate trade-off favoring accuracy and consistency over implementation simplicity:

Design Decision	Alternative Considered	Rationale
Hybrid Retrieval (Dense + BM25)	Dense-Only Vector Search	BM25 captures exact keyword matches that dense embeddings may miss, improving recall on technical terms.
Persistent Long-Term Memory	Session-Only Memory	Cross-session continuity enables personalized interactions and reduces repetitive context-setting.
Structured Persona Engineering	Simple System Prompting	Detailed persona modelling produces more consistent stylistic fidelity across diverse query types.
Retrieval + Cross-Encoder Reranking	Retrieval Alone	Reranking substantially improves precision by re-scoring candidates with full query context.

CONCLUSION

The Andrew Ng Digital Twin integrates four mutually reinforcing capabilities — Retrieval-Augmented Generation, persistent memory, structured persona engineering, and timeline-aware reasoning — into a unified conversational system. Each layer addresses a distinct failure mode: RAG combats hallucination, memory enables personalization, persona modelling ensures stylistic fidelity, and timeline reasoning prevents temporal inconsistency.

The result is a system that does not merely answer questions about Andrew Ng's work, but engages with users in a manner that authentically reflects his analytical rigor, pedagogical warmth, and practical orientation — grounded at every step in verifiable source material.